RURAL–URBAN DIFFERENCES IN STUDENT JOURNALISM PERFORMANCE METRICS*

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ABSTRACT

This paper examines rural-urban differences in student journalism performance among higher education institution (HEI) student-produced news outlets. With journalism quality frameworks, three article-level dimensions serve as comparisons: textual distinctiveness, Critical Information Needs (CIN) coverage, and localeness. Using a dataset of approximately 15,000 student articles, we utilized outlet-level soft cosine similarity, validated LLM-assisted binary coding, and logistic regression. Our results suggest three findings within our dataset: 1) the textual patterns of articles are not strongly structured by locale, 2) rural articles had substantially higher descriptive rates and odds of addressing a CIN than urban articles, and 3) rural articles had higher descriptive rates and odds of being coded as local. These results suggest that student journalism's local-news role varies more in their civic ability, corroborating that being locally engaged is not synchronous with addressing CINs and that rural-urban differences could extend beyond institutional capacities into HEIs informational role in surrounding communities.

1 Introduction

The difference between rural and urban areas can materialize in many different forms. One key area is the accessibility of information. Since, 2005, 80% of communities that lack credible sources of local journalism, coined as news deserts, are predominantly rural (Culpepper, 2025). This disproportionate relationship in information access between rural and urban areas can materialize into civic engagement and community well being, leading us to explore whether there are alternatives that can circumvent this inequality.

This paper explores whether student journalism can serve as a viable alternative to local journalism by comparing the quality of student journalism between urban and rural areas. We applied Napoli et al. (2017)'s conceptualization of quality to a series of comparative analysis involving cosine similarities to measure textual distinctiveness, LLM-assisted coding to assess outlet's ability to address critical information needs (CINs), and locale-based comparison to understand urban and rural student outlets' geographical coverage.

Research Questions

To understand whether the differences between rural and urban areas extend to student journalism, there were three questions that guided this paper:

- **RQ1:** To what extent to student outlets at rural and urban HEIs exhibit similar or distinct texts?
- **RQ2:** How do students outlets at rural and urban HEIs differ in the extent to which their articles address Critical Information Needs?
- **RQ3:** How do student outlets at rural and urban HEIs differ in their localness?

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2 Literature Review

The spread of information can be represented through information ecosystems, which are complex systems that include shared ideas, identities, institutions, social structures, technologies, people, and media outputs. In this system, information is both created and consumed by people with technologies and platforms that allow information to move (Wanless et al., 2025). However, access is not equal. Information access depends on whether information is available and trustable, which are shaped by norms, values, and communication habits within social groups (Burnett & Jaeger, 2008) and whenever there are information ecosystems where information is fragmented and unreliably accessible, information deserts form (Lee & Butler, 2019). Furthermore, information access is not only about whether that information exists or is accessible, but whether people perceive it as accessible (Culnan, 1985). Therefore, information inequality can be shaped through the organization of information in a community, and factors such as public policy, corporate control, media systems and technology can expand or restrict access to information (Burnett & Jaeger, 2008), potentially weakening local information infrastructure that then produces information deserts (Lee & Butler, 2019).

One crucial component of the ecosystem is local journalism. Some estimates that community newspapers provide up to 85% of "the news that feeds democracy" at both the state and local level (Jones, 2011), supporting observations that local newspaper abundance is associated with more accountable local government (Gao et al., 2020) and helping residents feel belonging to their community (Heiselberg & Hopmann, 2024). This is done through civic engagement as local news media coverage can positively affect voter turnout in local election (Backgaerd et al., 2014), counteract political polarization drive by national partisan media (Davis & Dunaway, 2016), and make citizens more able to evaluate their political leaders (Hayes & Lawless, 2015). Therefore, when local journalism becomes fragmented, they become news deserts, communities where access to local news and information is limited (Abernathy, 2023). In 2025, the U.S. has 213 counties with no local news source, up from 206 in 2024. Additionally, another 1,524 counties have only one remaining local news source, providing citizens limited access to reliable local news (Metzger, 2025). In those areas, residents lose both information access and community connection, which can isolate smaller neighborhoods from larger county-wide communities (Matthews, 2022). This also applies to Culnan's framework with residents who did not perceive their community as a news desert reporting stronger feelings of being informed and local issue knowledge (Collier, 2025). These effects can cascade through generations as young people in news-desert regions demonstrated signs of weaker news culture when they saw their localities only marginally represented in broader media, resulting in less self-initiated civic activties (Maneta et al., 2024), leading to "significant increases in property and victimless crimes, declines in voter turnout, and heightened political polarization" (Haddock et al., 2026).

There are a variety of theoretical reasons for the decline of local news. First, local news media receive lower audience traffic due to the digital economy (Hindman, 2018); in Portugal, some audiences increasingly prefer mobil devices and often do not want to pay for news (Quintanilha, 2018). As a result, local and regional newspaper struggle to balance adapting to digital consumption while preserving print products (Jenkin & Nielsen, 2020). After all, many local networks have experienced staffing reductions, recruitment challenges, long working ownership changes, and financialization where large financial institutions bought newspapers and prioritized short-term returns over historical missions, causing smaller communities to lose local newspapers (Abernathy, 2016). These dynamics are unevenly distributed across geographical and economical contexts. Communities that are economically struggling are more likely to lose a paper and lack local news alternatives (Abernathy, 2023). In the U.S., many news deserts concentrate in the Mountain States, the South, and the Great Plains. This creates a domino-effect where neighboring counties' local-news conditions can influence whether a county preserves their newspaper (Qin, 2026).

News deserts are also not just shaped by information access but their qualities. "Ghost papers" are newspapers that have become shells of their former serlves (Abernathy, 2018), fueled by corporate ownership and reduction in staff and resources that then leads to less original content and becoming more advertising oriented (Abernathy, 2020). Another variation are "zombie papers" which are papers that shrunk in size and frequency (Assmann, 2023). Even then, local-news scarcity can be measured in several ways. First, fundamentally, scarcity can be measured through availability of local news across communities (Abernathy, 2018; Abernathy, 2020). Locality is especially important since public radio broadcasts do not necessarily equal local news. Second, news deserts can also emerge from journalistic labor deficits even when media infrastructure will exist (Olsen & Mathisen, 2023). Third, even if infrastructure and staff is preserved, news deserts can also be measured at the content level. For instance, geographic diversity in news content is a factor, which includes continental and domestic levels (Volger et al., 2023; Madrid-Morasles et al., 2023). In this paper, we consider rurality as an important level.

Rural areas consist of communities that face broad structural challenges across resources, infrastructure, employment, education, and others (Helpap, 2023; Mueller et al., 2021). However, they are not necessarily homogenous, as they vary by economic base, leading to limited public service accessibility (Lofaro & Schwind, 2026), demographic composition, political culture, and social norms (Roger, 2019), and these socioeconomic conditions such as government disinvestment can shape rural political attitudes with public institutions (Cramer, 2016; Hochschild, 2016).

Contextually, this forms divisions between geographics, particularly between rural and urban areas. Practically, whenever policies are designed from urban evidence, they struggle to account for rural contexts which requires special attention to the conditions of rurality (Brunet, 2015; Helpap, 2023). This ties in with rural Americans generally having differing political and culture views from their urban counterparts (Gimpel & Karnes, 2006). And while technological innovation can reduce rural disadvantages, these benefits only appear with communities of sufficient connectivity and resources, which widens the gap between connected and isolated rural areas (Johnson, 2001). Rural areas also have structurally higher public service costs than urban areas (Shani et al., 2025).

These differences fuel disproportionate information inaccessibility between rural and urban areas. Since 2005, 80% of all news deserts are predominantly rural (Culpepper, 2025). Libraries, a major source of information, lag behind urban libraries in technology, staff, and funding (Real, 2014). Even then, media sources favor central areas and leave peripheral areas isolated, even when those areas have necessary infrastructure, indicating that financial resources do not necessarily correlate with media presence (Saiz-Echezarreta et al., 2026). Therefore, local communication channels remain an integral part for rural civic engagement and to combat the nationalization of politics (Thompson, 2021), stressing the importance of public funding to improve the abilities of these organizations (Neff et al., 2022).

Aside from local news organizations, another component of information ecosystems are higher education institutions (HEIs). HEIs can act as an anchor institutions where they convert their assets into place-based public capacities such as employment, land, research, staff expertise, and infrastructure (Jeffrey, 2025). Such mechanisms are collaboration with media partners for support (Hutchins, 2025) and student journalism. Once utilized to purely inform on campus-level topics (Hapney & Russo, 2013), roughly 70% of student media organizations engage in community reporting activities (Allen & Bhowmik, 2025), which is done through partnering with local news organizations (Bhowmik, 2025) and placing greater emphasis on the service role of journalism in news deserts (Finneman et al., 2022).

When it comes to rural HEIs and specifically rural student newspapers, more unique functions arise. In rural areas, regional HEIs become deeply embedded in their communities (Supple & Orphan, 2023), becoming major employers in their communities (McClure et al., 2021) and resouce hubs that provide critical services where access is limited (Norris et al., 2023). However, despite their functions, around 20% of agency leaders report that their state has specific funding provisions for rural HEIs (Koricich et al., 2020), possibly due to smaller HEIs being structurally disadvantaged in a marketized higher education system where they lack prestige, selectivity, and pricing power for their counterparts (Kim et al., 2025). Beyond government funding, endowment distribution is unequal, with the top 20% of HEIs holding 80% of the total wealth and the bottom 20% holding around 1% (Day & Norton, 2023).

At least 56% of student journalism organizations have major financial ties to their respective HEIs (Sparks et al., 2024), which allows cases of defunding for these organizations (American Association of University Professors, 2016), indicating that funding is an important component to the health of student journalism (Salahi & Smith, 2022). Considering the financial inequalities smaller HEIs face, it becomes possible that these discrepancies can materialize within student journalism.

Therefore, the purpose of this paper is to study whether student outlets differ across rural and urban contexts in the kind of coverage they produce. It then becomes necessary to operationalize quality through measurable article-level dimensions. Some works emphasizes the geographical relevance, depth and breadth of reporting, and presentation qualities as measurable outcomes (Lacy & Rosenstiel, 2015), yet in order to account emphasis of the existence of local news outlet as essential to democratic life, we have to consider whether the content is relevant to communities (Usher, 2023). Ultimately, Napoli et al. (2017) provides a methodological framework that accounts for "infrastructure (the availability of journalistic sources), output (the quantity of journalistic output from these sources), and performance (the extent to which this output is original, is about the local community, and addresses critical information needs)." We specifically are interested in performance, so we operationalize the quality of student journalism in three components: textual distinctiveness, ability to address CIN, and geographic coverage.

Given student journalism's potential to report and assess community-level news due to their embedded nature within a community (Cowan & Foster, 2024), adapting the assessment to their ability to address CINs fundamental to both individuals and communities (Friendland et al., 2012) allows us to operationalize student journalism while accounting for the fluidity of news media and its relationships to a deliberative democracy (Bachmann et al., 2022).

Additionally, diversity of news coverage includes geographical diversity (Humprecht & Esser, 2018), meaning news media should address a variety of places (Volger et al., 2023). Since local news stations acquired by large corporations may report more frequently on national news at the expense of local topics (Wanner et al., 2025), such upscaling can threaten geographical diversity (Vogler et al., 2023). Therefore, assessment of student journalism and their geographical coverage furthers understanding of student journalism's ability to assess declines in local news coverage.

3 Methodology

3.1 Theoretical Framework

Building on the literature, the paper treats student-produced news outlets as part of local information ecosystems attached to HEIs. In order to quantitatively compare urban and rural outlets, we define the geographic comparisons group and operationalize journalistic performance through textual distinctiveness, CIN coverage, and geographic coverage.

Rurality. To define what is considered rural and urban, we operationalize rurality through the National Center of Education Statistics' IPEDS locale framework. To account for the varying levels of urbanization and rurality, we adopted Jenks et al. (2025)'s method of collapsing IPEDS town/rural designations into rural and city/suburban into urban. Therefore, in our paper, rural HEIs consist of locale codes 31, 32, 33, 41, 42, 43, and urban HEIs consist of locale codes 11, 12, 13, 21, 22, 23.

Our dataset includes a variety of student journalism outlets such as traditional student newspaper, journalism-school news services, local reporting projects, statehouse bureaus, and specialized reporting projects. We do not assume that each outlet in the dataset is the university's only student outlet and consist of the same editorial mission. Rather, we define these organizations as observable, university-affiliated student-produced news outlets.

Operationalization. To account for journalism quality's many possible operational dimensions (Lacy & Rosenstiel, 2015), we adopt Napoli et al. 2017's framework. We are specifically interested in performance, so we operationalize the quality of student journalism in three components: originality, ability to address CINs, and geographic coverage.

Originality. Napoli's originality concept asks whether a story provides a novel contribution or republishes information. However, this requires comparing every article to a network of other possible sources. Instead, we evaluate whether the overall publication corpus of every outlet is textually distinctive between rurality and urbanism. We consider this more exploratory rather than a direct form of the originality measure.

CIN. To operationalize the ability to adddress CIN, we explore whether the article addresses a CIN through a dichotomous variable, similar to Napoli et al. (2018)'s approach.

Locality. We adopt Napoli et al. (2017)'s specific geographical focus, where an article is to be local if the subject was oriented around a specific town or a smaller geographic unit. This includes neighborhood, campus community, or local institution.

3.2 Quantitative Framework

Data

This paper uses a dataset of university-affiliated student produced news articles provided by the Center for Community News (Cooley, 2026). The outlets in this dataset were collected from StudentMediaMap, a database of more than 1,000 identified student media outlets. The dataset was constructed at the outlet level, where the data collection process began with the student media outlets and then identified the HEI associated with each outlet. Through a web-scraping process, article-level title, author, and body text was extracted.

Determining Rurality

With the dataset, we linked each outlet to the associated HEI through manual processing and LLM verification and created an unique list of HEIs. then, we utilized the 2024 IPEDS institutional directory file containing the variables UNITID, INSTNM, LOCALE to match each HEI in the dataset to its corresponding IPEDS institution through exact and fuzzy string matching with manual validation. Afterwards, we are able to convert each locale code into a geographic group with local codes, where we were able to categorize each article and outlet to a binary rurality value.

Originality

To evaluate whether rural and urban student outlets produce distinct textual portfolios (RQ1), we compared each outlet to every other outlet using soft cosine similarity measures (Sidorov et al., 2014), where we compared the overall body of coverage produced by each outlet rather than individual articles.

Outlet-level corpora were constructed from article bodies only. To prevent corruption of vectorization, non-story metadata were excluded before analysis. Each corpus was tokenized with Gensim's preprocessing tools, lowercased, stripped of punctuation and accent marks, and filtered to remove stop words and tokens shorter than three characters. To

reduce noise and improve computational efficiency, the dictionary was filtered to exclude tokens appear in fewer than five corpora or in more than 60% of corpora, and the vocabulary was limited to the 10,000 most frequent remaining terms. The cleaned corpora were represented as TF-IDF weighted vectors, and soft cosine similarity was calculated using glove-wiki-gigiaword-50 and pretrained GloVe word embeddings (Pennington et al., 2014).

This was done for every unique pair of publication corpus, and each pair was classified according to the geographic status of the two outlets: within rural (n=17,706), within-urban (n=214,840), or between-group (n=123,984).

However, because the number of these pairs differed substantially, we conducted an equal-sized bootstrap (Efron, 1979) to estimate the stability of our previous results. For 5,000 iterations, we sampled 17,766 pairs, corresponding to the number of available within-rural pairs, with replacement from each pair type and recalculated group mean similarity scores.

LLM-assisted Coding Procedures

The unit of analysis is individual articles of each outlet, where each articles receives a CIN and local binary indicator. To generate these indicators, we applied a LLM-assisted framework due to its flexibility (Zupic, 2026) and ability to perform similar to supervised machine learning and replicate traditional qualitative methods (Than et al., 2025).

We applied Zupic (2026)'s framework which involves giving a LLM a codebook, instructions, and few shot prompting to improve accuracy (Carlson & Burbano, 2025) given through API processing to analyze large corpora (Churchill et al., 2025). The codebook was created with a LLM workflow, which is found to produce codebooks with a larger number of relevant codes and higher overall quality (Barany et al., 2024), modeled after Zambrano et al. (2026)'s approach. This process includes providing the model with a theoretical framework and evaluating its candidate codes/examples with human review.

Our evaluation/validation process includes creating a development (n=25) and evaluation (n=50) set that are sampled independently from the broader corpus to avoid overlap (Lin & Zhang, 2025), with our development set responsible for improving the codebook and the evaluation set to confirm with LLM's outputs are reliable (Abdurahman et al., 2025) by ensuring that the labels match with the model's results (Karjus, 2023).

We model our validation thresholds after Törnberg (2024), which includes using a sufficiently representative validation set, task-appropriate performance metrics, and benchmarking against human coders.

With our first iteration of the CIN codebook, we manually (n=2) reviewed the development set, reported kappa values, and resolved conflicts to establish golden values. then, we had gpt-5.5 code the development set, and we compared the model's values with the golden values to report our evaluation metrics: F1, specificity, and kappa as robustness checks. We resolved any model-human conflicts and further developed the codebook.

We used the resolved codebook to manually code the evaluation set. Following the same procedures with the development set, we reported kappa values and established golden values. Then, we had gpt-5.5, gpt-5.4, gpt-5.4-mini, and gpt-5.4-nano undergo similar procedures and reported F1, specificity, and kappa.

CIN Coverage

To assess whether articles from rural outlets are more or less likely to address a CIN than articles from urban outlets, our first codebook had three inclusion standards: the article had to substantively address a CIN topic; the article had to provide information beyond local campus life; the CIN issue had to be developed enough to constitute as one of the article's main points. We calculated an initial human-huamn agreement at 84.0% with $\kappa = 0.655$.

Then, the revised codebook was implemented in the LLM prompt, and the LLM's classifications were compared with the golden values. Eventually, to optimize price-efficiency, our model of choice is gpt-5.4-mini, a smaller model that is more useful for binary classifications (Zupic, 2026).

We then estimated a logistic regression model. The dependent variable is the binary CIN indicator, and the independent variable is the binary rural indicator. Because articles are nested within the outlet, standard errors are clustered at the outlet level. The model is specified as:

$$\text{logit}\left[\Pr(CIN_{ij} = 1)\right] = \beta_0 + \beta_1 \text{TownRural}_j + \epsilon_{ij}$$

where i indexes articles and j indexes outlets. CIN_{ij} indicates whether article i from outlet j addresses a critical information need, and TownRural_j indicates whether outlet j is affiliated with a rural HEI. The coefficient β_1 estimates whether articles from rural outlets differ from articles from their urban counterparts in their likelihood of addressing a CIN. Results are reported as predicted probabilities.

Locality

We applied a similar LLM-assisted classifer to assign a binary local indicator to article text, and similar validation procedures were conducted.

We first estimate a logistic regression model using the full article-level dataset. The dependent variable is the binary locality indicator, $Local_{ij}$, and the independent variable is the binary rural indicator, $TownRural_j$. Because articles are nested within outlets, standard errors are clustered at the outlet level. The model is specified as:

$$\text{logit}\left[\Pr(Local_{ij} = 1)\right] = \beta_0 + \beta_1 TownRural_j + \epsilon_{ij}$$

where i indexes articles and j indexes outlets. $Local_{ij}$ indicates whether article i from outlet j is classified as local, while $TownRural_j$ indicates whether outlet j is affiliated with a rural HEI. The coefficient β_1 estimates whether articles from rural outlets differ from articles from urban outlets in their likelihood of producing local coverage. Results are reported as predicted probabilities.

4 Results

Validation of LLM-assisted Coding

Table 1: *CIN Classification*

Metric	Human	GPT-5.5	GPT-5.4 Nano	GPT-5.4 Mini	GPT-5.4
Agreement/Accuracy	0.900	0.880	0.760	0.800	0.800
Precision	–	0.857	0.929	0.882	0.933
Recall/Sensitivity	–	1.000	0.722	0.833	0.778
F1 Score	–	0.923	0.813	0.857	0.848
Specificity	–	0.571	0.857	0.714	0.857
Cohen's Kappa	0.746	0.658	0.493	0.548	0.552
Balanced Accuracy	–	0.786	0.790	0.774	0.817

Note. This table reports human-coder reliability and LLM validation results for the locality classification task. The purpose of the table is to assess whether the locality codebook could be applied consistently by human coders and whether LLM-assisted classification produced results reliable enough for use on the broader article dataset. Two evaluation rounds are reported because the first round was used as a development evaluation. Results from the first evaluation indicated that the initial codebook and prompt required additional clarification, so disagreement patterns were reviewed and used to refine the classification rules. After those revisions, a second validation evaluation was conducted on a separate sample. The second evaluation is therefore treated as the primary post-revision validation of the locality classification procedure.

Table 1 indicate that human coders achieved a strong agreement on the CIN classification task ($\kappa = 0.746$). Among the models, we found that while GPT-5.5 performed best overall ($F_1 = 0.923$), its lower specificity score (0.571) suggest that a it was more likely to classify some non-CIN articles as CIN. Therefore, we opted for GPT-5.4 Mini that provides a more balanced distribution between F1 and specificity.

Table 2: *Locality Classification*

	Development Evaluation		**Validation Evaluation**	
Metric	Human	GPT-5.4 Mini	Human	GPT-5.4 Mini
Agreement/Accuracy	0.860	0.800	0.940	0.840
Precision	-	0.786	-	0.848
Recall/Sensitivity	-	0.611	-	0.903
F1 Score	-	0.688	-	0.875
Specificity	-	0.906	-	0.737
Cohen's Kappa	0.692	0.544	0.874	0.653
Balanced Accuracy	-	0.759	-	0.820

Note. This table reports human-coder reliability and LLM validation results for the locality classification task, with the goal of assessing whether the codebook could be applied consistently and whether LLM-assisted classification was reliable enough for use on the broader article dataset. Two evaluation rounds are included because the first served as a development evaluation: its results revealed areas where the codebook and prompt needed clarification, leading to revisions before a second, separate validation evaluation was conducted and treated as the primary post-revision assessment.

During our locality classification process, the first round, labeled as development evaluation in Table 2, was originally intended to serve as the validation evaluation, but its results indicated that the locality codebook required additional work. After revisions, another validation evaluation was conducted. Indicated in Table 2, human and model reliability improved ($\kappa = 0.874$; $F_1 = 0.875$).

Cosine Similarity Analysis

Table 3: *Original Pairwise Soft Cosine Similarity*

Pair type	N pairs	Mean	Median	SD	Min	Max
Rural–rural	17,766	0.381	0.381	0.100	0.000	0.896
Rural–urban	123,984	0.377	0.378	0.104	0.000	1.000
Urban–urban	214,840	0.375	0.378	0.109	0.000	1.000

Note. This table reports the original pairwise soft cosine similarity values for publication-level corpora, with number of unique outlet pairs in each group, mean, median, standard deviation, and minimum/maximum values to summarize the distribution. Higher values indicate greater textual similarity between publication-level corpora.

Table 4: *Bootstrapped Soft Cosine Similarity Estimates*

Pair type	Mean	Median	95% CI
Rural–rural	0.381	0.381	[0.379, 0.382]
Rural–urban	0.377	0.377	[0.375, 0.378]
Urban–urban	0.375	0.375	[0.374, 0.377]

Note. This table reports the bootstrapped soft cosine similarity estimates by outlet pair type. Bootstrap estimates are based on 5,000 iterations, with 17,766 pairwise comparisons sampled from each geographic pair type to equalize comparison-group size. The 95% confidence intervals are based on the distribution of bootstrapped mean similarity estimates. Higher values indicate greater textual similarity between publication-level corpora.

Tables 1 and 2 report soft cosine similarities for the original and bootstrapped analyses. After accounting for the unequal number of pairs, we found minimal change in the mean soft cosine similarity values. Therefore, the observed means likely reflect differences in the corpora of rural and urban outlets, with little effect from differences in pair-group size.

In regards to textual distinctiveness, since all three pair types' similarity scores are clustered together, this suggest that there are no large differences in textual distinctiveness by geographical outlet type.

CIN

Table 5: *Descriptive Results: CIN*

Type	Articles	Outlets	CIN Articles	CIN Rate
Urban	10,702	654	2,455	0.2294
Rural	2,570	188	353	0.1374

Note. This table reports the number of articles, number of outlets, number of articles coded as addressing a CIN, and the CIN rate for urban and town/rural outlets.

Table 5 indicates that urban articles were more likely than rural articles to address a CIN. Therefore, CIN coverage is more common among urban outlets than rural outlets.

Table 6: *Logistic Regression Predicting CIN Coverage*

Model	Variable	Coefficient	SE	Odds Ratio	95% CI	p
Unclustered	Constant	-1.212	0.023	0.298	[0.285, 0.311]	< 0.001
	Rural	-0.626	0.062	0.535	[0.474, 0.604]	< 0.001
Clustered	Constant	-1.212	0.077	0.298	[0.256, 0.346]	< 0.001
	Rural	-0.626	0.163	0.535	[0.389, 0.736]	< 0.001

Note. The dependent variable is *final_CIN_binary*. The constant reports the estimated baseline odds of CIN coverage for urban articles. The coefficient reports the direction and size of the relationship between rural status and CIN coverage. The standard error reports the uncertainty around that estimate. The odds ratio shows how the odds of addressing a CIN differ for rural articles compared with urban articles. The 95% confidence interval reports the range of plausible odds-ratio values, and the p-value indicates whether the estimated relationship is statistically significant. The outlet-clustered model adjusts standard errors to account for articles being grouped within the same outlet.

Table 6 indicates that rural status is negatively associated with CIN. In the unclustered and clustered model, the rural coefficient is -0.626, indicating that rural articles have a lower log-odds of addressing a CIN than urban articles. The odds ratio of 0.535 indicates that the odds for rural articles to address a CIN are about 46.5% than urban articles.

This relationship being statistically significant after clustering standard errors suggest that the lower rate of CIN coverage initially found is both descriptive and statistically supported.

Locality

Table 7: *Descriptive Results: Locality*

Type	Articles	Outlets	CIN Articles	CIN Rate
Urban	10,706	654	5,544	0.5178
Rural	2,571	188	1,468	0.5710

Note. This table reports the number of articles, number of outlets, number of articles coded as local, and the locality rate for urban and town/rural outlets.

Table 7 indicates that locality is minimally more common among rural outlets than rural outlets as rural articles were more likely than urban articles to be coded as local.

Table 8: *Logistic Regression Predicting Locality*

Model	Variable	Coefficient	SE	Odds Ratio	95% CI	p
Unclustered	Constant	0.071	0.019	1.074	[1.034, 1.115]	< 0.001
	Rural	0.214	0.044	1.239	[1.1365, 1.352]	< 0.001
Clustered	Constant	0.071	0.053	1.074	[0.968, 1.192]	0.179
	Rural	0.214	0.119	1.239	[0.982, 1.564]	0.071

Note. The dependent variable is *final_locality_binary*. The constant reports the estimated baseline odds of locality for urban articles. The coefficient reports the direction and size of the relationship between rural status and locality. The standard error reports the uncertainty around that estimate. The odds ratio shows how the odds of locality differ for rural articles compared with urban articles. The 95% confidence interval reports the range of plausible odds-ratio values, and the p-value indicates whether the estimated relationship is statistically significant. The outlet-clustered model adjusts standard errors to account for articles being grouped within the same outlet.

Table 8 does demonstrate that rural status is positively associated with locality. Both the unclustered and cluster models had their rural coefficient as 0.214, indicating that rural articles had a higher log-odds of being coded as local than urban articles. The odds ratio being 1.239 means that the odds of locality for rural are about 23.9% higher than odds for urban articles.

However, after clustering, the relationship is no longer statistically significant with the p-value being off of the conventional level slightly ($p = 0.071$). This suggests that the descriptive difference is positive and should be interpreted cautiously.

5 Discussion

5.1 Research Questions

RQ1: To what extent do student outlets at rural and urban HEIs exhibit similar or distinct texts?

In our cosine similarity analysis, we assumed that if locale did strongly structure journalistic form through textual patterns, then articles from the same locale category would be expected to resemble each other more than from different locale categories. Therefore, we expect the rural-urban pair type to have a substantially lower similarity score than the two within-group pair types.

However, we found that the three pair-type values were extremely close to each other, where the rural-urban pair type is not meaningfully different from the other two within-group values. From that, we determine that within our dataset, the corpora were not organized into clearly separate rural and urban textual separations.

Broadly, it appears that locale does not strongly structure the textual similarity patterns of our dataset. A content analysis of the top six U.S. student newspapers found that those newspapers exhibit consistent patterns of topic selection, emphasizing campus administration, student government, athletics, and campus events (Schmidt, 2014). Our results introduces nuance. Our soft cosine values indicate that, on average, every article, compared to another article, is mostly moderately different given the confidence intervals. However, the confidence intervals do stay similar when adjusted for different pair types. this indicates that the locale of our articles in our dataset does not correlate with changes in the soft cosine values.

Following the analysis, we used a 3D UMAP visualization as a visual check of whether the rural and urban articles did not occupy distinct semantic regions.

Figure 1 shows substantial overlap across the rural and urban articles, where we can observe that rural and urban articles are dispersed throughout many of the same clusters. This is consistent with the soft cosine similarity results.

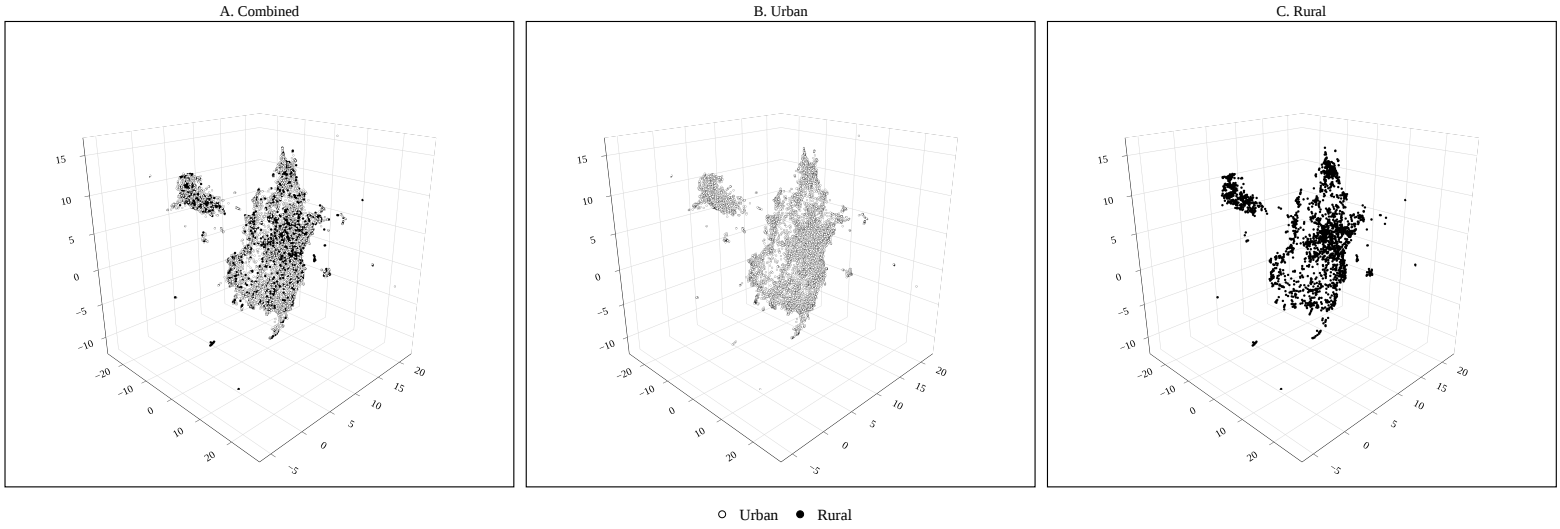


Figure 1: Article-Level 3D UMAP Projection by Rural/Urban Status

RQ2: How do student outlets at rural and urban HEIs differ in the extent to which their articles addressed CINs?

From our regression results, we found that within our dataset, rural articles had substantially lower odds of addressing a CIN than urban articles. This is a result that remained statistically significant after clustering. Additionally, our descriptive results demonstrated that rural articles had a lower CIN rate than urban articles.

Our results are consistent with the theoretical possibility that urban outlets have a greater capacity CIN-related coverage. Prior literature found that the production of information addressing CIN depends heavily on the capacity of local news organizations (Waldman & Working Group on Information Needs of Communities, 2011) and specifically connected

it to rural communities, finding that rural newspapers face higher understaffing, causing them to struggle to cover local news (Abernathy, 2016). Descriptively, our results match with the finding that 80% of all news deserts being predominantly rural (Culpepper, 2025) and expanded this understanding by specifically connecting these dynamics towards student journalism.

RQ3: How do students at rural and urban HEIs differ in their geographic coverage?

From our regression results, we found that within our dataset, there is limited statistical significance in whether rural articles have higher odds in addressing a local issue than urban articles. Yet, descriptively, we found that rural articles do have a higher rate of being coded as local. Nagel and Broersma (2026), through a fieldwork of rural communities, found that their respective newspapers acted as community boosters and felt obligated to advance local interests.

Our descriptive results provided quantitative support for that dynamic, specifically within student journalism. While our clustered model does not meet the conventional p-value thresholds, we argue that it is still possible that within our dataset, rural articles have higher odds of being coded as local versus urban articles.

5.2 Broader Implications

Student Journalism and Addressing Local Needs

The Center for Community News found approximately 70% of student media organizations have engaged in some form of community reporting activities, with 57% expressing desire to cover more community-related topics (Allen & Bhowmik, 2025). Our paper contextualizes this impact through looking at student journalism's potential on addressing CIN and maintainng locality.

In regards to CIN, we found that both urban and rural articles are limited to their ability to address a CIN. In our dataset, we found that 13.7% of rural articles and 22.9% of urban articles addressed a CIN. Concurrently, we found that 57.1% of rural articles and 51.8% of urban articles were coded as local.

For student reporting, this introduces an important finding: reporting on local topics does not necessarily correlate with reporting on CIN. Within our dataset, some of the articles that were coded as local reported on topics such as local events, campus stories, and other local-related topics. Yet, to be coded as CIN, articles had to be more restricted to a taxonomy. Therefore, we provided experimental evidence why in Napoli et al. (2017)'s framework, locality and CIN should be measured separately and not considered synchronous, especially in the context of student journalism.

Rural-Urban Dynamics

Earlier, we theorized that "considering the financial inequalities that smaller HEIs face, it becomes possible that these discrepancies can materialize within student journalism." Our quantitative results demonstrate that there is a connection between institutional-level inequality to article-level journalism metrics, where we further contextualize the literature on rural-urban inequalities by connecting these resources inequalities to its implications in student journalism and its performance metrics.

Specifically, our CIN analysis indicating that urban outlets have higher odds of CIN coverage suggests that on aggregate, urban outlets, which generally operate in larger and institutionally denser environments, would have better access to resources that contributes to their ability to inform on CIN.

Interestingly, the unequal resource dynamic does not apply to our local analysis, where we found that rural outlets have higher odds to be more locally attached. This corroborates with current literature that finds that rural HEIs, within their community may be a more major local institution embedding relative to urban institutions and their communities. Our findings extend this framework into student journalism and its connection to HEIs.

The unrelated relationship between locality and ability to address CIN indicates that whiel rural HEIs may be more connected to their community, they are not necessarily providing information on CIN.

5.3 Limitations

Dataset

There are a variety of issues within the dataset that can affect our full results. First, there are uneven corpus sizes between rural and urban. Our final cleaned dataset has 654 urban outlets with a total of 10,702 articles and 188 rural outlets with a total of 2,570 articles. This uneven distribution undoubtedly skews our analysis. Since the urban corpus has more articles and outlets, its model estimates are more stable than the rural corpus, where a few unstable outlets

could influence the overall pattern. This explains why our rural estimates have larger standard errors and confidence intervals. Therefore, while our findings can support claims about articles in the current version of the dataset, there needs to be more caution where generalizing to all rural and urban student outlets.

Second, the dataset includes different types of student outlets. Currently, the dataset is primarily built on two types: student newspapers and news-academic partnership. The primary difference is that they operate under different editorial structures, where student newspapers often operate with student editors and news-academic partnerships involve journalism schools, faculty, and other professional editors. This fundamentally affects the content of their articles, potentially materializing our observed differences in our model.

Third, the timelines of our articles in the dataset are different in publication rate and consistency. Some of the outlets publish daily or weekly, with others publishing more irregularly. This affects outlet-level consistency, where we see some outlets have high amounts of articles and others with low amounts. This is also affected by the RSS feed availability. Scraping can be affected across outlets, with some outlets providing complete or truncated article feeds. Therefore, outlets with more favorable RSS feeds may be overrepresented over outlets with less favorable feeds. this affects our clustered models. Some outlets may contribute to a different number of articles, clustering will likely also reflect those uneven distributions, where we have some outlets properly represented and some having too few articles to represent their typical coverage behavior.

Classification of rural

In our paper, we followed Jenks et al. (2025)'s approach in collapsing IPEDS town and rural designations. We also had a statistical reason for such. If we were to classify articles on a four-point-spectrum, we would have 48 rural outlets, 147 town outlets, 203 suburban outlets, and 460 city outlets. Nevertheless, such a decision eliminates any analysis of the specific nuances between the four designations. Therefore, on the theoretical level, our findings may not reflect ture rural-urban differences.

Soft cosine and originality

Our soft cosine similarity measures textual similarity, and it does not tell us about the originality of the article such as whether a story was first reported by the student outlet. This is an ongoing issue, where we struggled to consider methods to measure an article's originality on a larger scale. Therefore, our soft cosine results are framed in terms of textual distinctiveness between outlet rather than their levels of originality.

Binary coding

For the logistic regression, we had to simplify complex journalistic concepts into binary indicators. However, this limits our ability to capture nuance. For instance, while an article can be coded as either addressing or not addressing a CIN, the binary measure does not capture the extent, depth, or quality of which the article did address a CIN.

For localness, the binary measure does not distinguish among different levels of types of localness, such as campus-level or town-level. Specifically, while our findings did see a higher rate for rural articles to be coded as local, this does not automatically mean they are focused heavily on the surrounding town community. For example it is possible they are primarily campus-level.

Therefore, the differences we report on CIN coverage and localness should be interpreted as whether the characteristic was present, but not in degrees or what forms it has appeared.

5.4 Future Directions

Rurality and greater specificity

Future research should represent rurality with greater geographic precision relying only a broad rural and urban binary spectrum. While this conceptual decision was useful to create a clean comparison, there are many important geographic variations. One clear variation is the distinction between town and rural. For example, a college in a small town could have different levels of institutional resources compared to a remote rural area. For the urban group, suburban HEIs likely do not operate in the same local-news environment as heavily dense city-based areas, where city-based HEIs may have more access to professional newsrooms, civic institutions, etc.

Therefore, one approach could be to examine the full IPEDS locale codes separately, where outlets would be classified among these four codes. Another approach is to distinguish between fringe, distance, and remote classifications within each locale designation. Beyond IPEDS, another approach would be to use alternative measures of rurality, where it allows future studies to examine student jounralism across degrees of rurality rather than a single binary condition.

A scalable aggregate measure of originality

There should be a more direct and scalable measure of journalistic originality. Some possible indicators could be the pressure of named human sources, number of quotes sources, diversity of source types, or similiarity to professional local-news coverage. Another approach is to create an index that combines several indicators. As mentioned before, the primary issue is developing a measure that can be properly scaled to a large corpora.

Dimensionalize CIN and locality

Future research should develop a more nuanced measure of CIN and localness. Our current binary approach is useful for large-scale classification and statistical analysis, but utilizing multidimensional measures allows for more nuanced research findings. For CIN, some dimensions could be CIN category, depth of coverage, or source diversity. For example, exploring the CIN category can potentially expose hidden nuances between different types of coverage, since not all CIN serve the same function. Level of depth can introduce nuance to the level of public value that a piece of news article can provide, which can be measured through source diversity.

For localness, breaking the level of localness into campus, town or city, county, etc would clarify what aspects of "local" student outlets are actually covering. This allows future exploratin of whether these outlets are covering capus issues or surrounding communities.

6 Conclusion

In this paper, we have conceptualized and measured the performance of articles from student journalism outlets. Through this, we were able to compare articles affiliated with rural and urban HEIs and explore differences in textual distinctiveness, ability to address a CIN, and localness.

Our dataset consisted of over 13,000 articles scraped from over 800 student outlets that consist of standard student newspapers and news-academic partnerships. To measure and compare textual distinctiveness, each outlet-level corpus is vectorized with TF-IDF, and pre-trained GloVe embeddings were utilized to calculate soft-cosine values. For both ability to address a CIN and localness, we collapsed each category as a dichotomous value and utilized a LLM-pipeline to code each article for both CIN and localness. To compare, we used logistic regression with rurality as the predictor and CIN and localness as the outcomes.

Within our dataset, we found evidence that rurality does not appear to strongly impact textual distinctiveness. We also found that the rural articles were less likely to be coded as addressing a CIN ($p < 0.001$) and, cautiously, were more likely to be coded as local ($p = 0.071$).

We then contextualized these results in rural-urban inequalities and student journalism's potential in the face of news deserts.

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